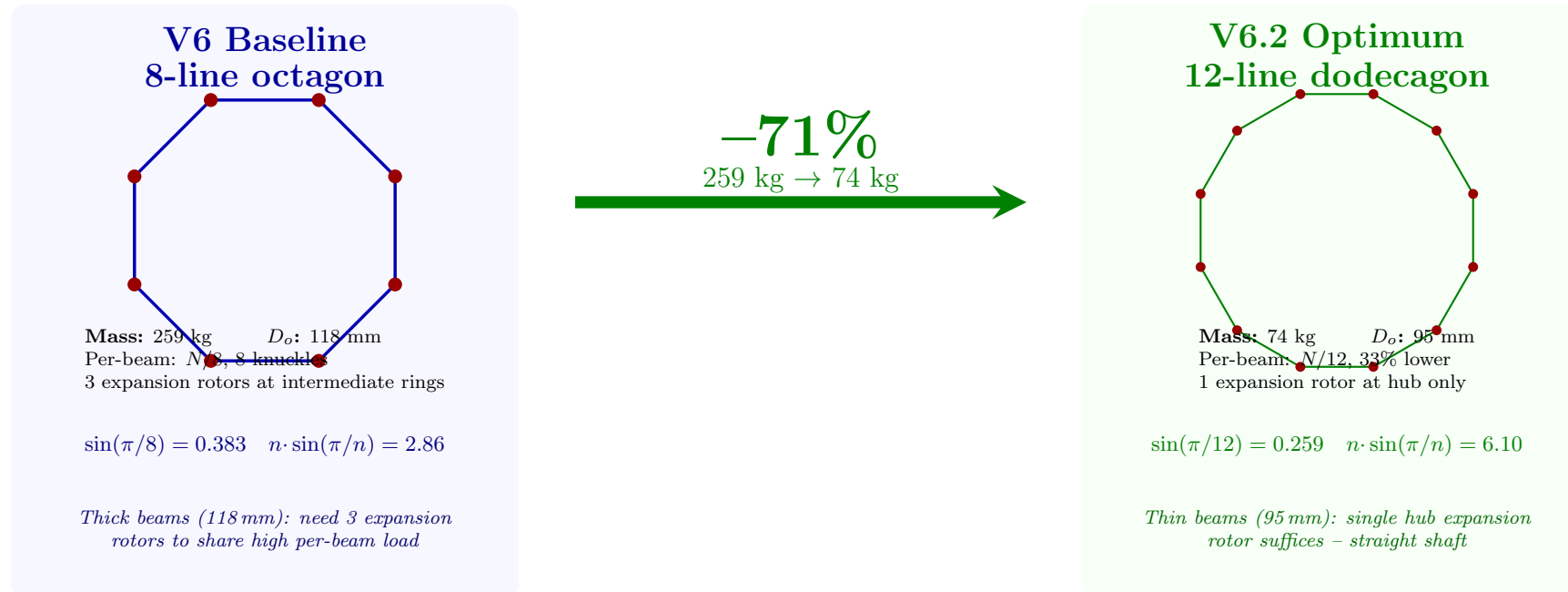


The Polygon Flip: Why 12 Lines Beat 8

Beam-only scaling says fewer lines are lighter. The DE optimiser found the opposite – 71% mass reduction.



What this reveals about the TRPT scaling problem:

Three factors drove the 71% mass reduction (259→74 kg), not just the polygon change:

- 1. Coupled knuckle mass:** The old model used fixed 0.005 kg knuckles regardless of beam diameter. The corrected model derives knuckle mass from beam geometry – 118 mm beams cost far more in knuckle weight than 95 mm beams, exposing the true cost of the V6 baseline.
- 2. Expansion rotor consolidation:** V6 needed 3 expansion rotors at intermediate rings to share the high per-beam load ($N/8$ compression). At $n=12$, per-beam compression drops to $N/12$ – 4× lower – so a single hub expansion rotor is sufficient. Going from 3 rotors to 1 saves substantial blade and hardware mass.
- 3. Corrected physics:** The $\tan \rightarrow \sin$ polygon force correction and $\cos^3 \rightarrow \cos^{2.65}$ elevation correction eliminated artificial safety margins that had hidden the true structural cost.